

Markscheme

Mathematics: analysis and approaches
Practice question bank

105. (a)

It can be proven by contradiction that there are no positive integers x and y satisfying $x^2 - y^2 = 2$ (factorise as $(x - y)(x + y) = 2$, and show no pair of positive integer factors of 2 gives integer solutions for x and y).

Using $x = 6$ and $y = 4$ (which do not satisfy the equation above), find the value of $x^2 - y^2$, and confirm it is not equal to 2.

$$x^2 - y^2 = 36 - 16 \quad (M1)$$

$= 20$, which is not equal to 2, consistent with the proof A1

[6 marks]